

2024, Mar. 2024 Release LOC:Acute Adult

Infection: CNS

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review ^(1, 2, 3, 4)Episode Day 1 ^(1, 3, 4)Episode Day 2 ^(3, 4, 46)Episode Day 3-7 ^(4, 59, 60)

Episode Day 8

Discharge Screens ⁽⁹⁴⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for patients with central nervous system (CNS) infections including meningitis, encephalitis, and brain abscess. For patients who begin to exhibit signs and symptoms consistent with sepsis and need additional interventions which are not provided in the current subset, please refer to the Infection: Sepsis subset.

Meningitis: Meningitis is an inflammation of the membranes surrounding the brain and spinal cord, most often caused by infection. A suspicion of meningitis is based on the patient's clinical symptoms which may include headache, fever, neck stiffness, and altered mental status. In older adults, a change in level of consciousness may be the only sign (Magazzini et al., Intern Emerg Med 2012, 7: 359-64). Rapid differentiation between acute bacterial and aseptic (e.g., viral, fungal, tuberculosis (TB), Lyme) meningitis, and prompt initiation of treatment targeting the offending pathogen are key to optimizing patient outcomes. Diagnosis is based on a thorough history, physical examination, blood culture results, and cerebral spinal fluid (CSF) analysis. CSF should undergo glucose, protein, and cell count analysis as well as culture and sensitivity testing. Rapid diagnostic tests, such as latex agglutination and polymerase chain reaction (PCR), may also be used in conjunction with culture and sensitivity testing depending upon local availability. PCR testing is particularly useful due to its high sensitivity and specificity for pathogen detection (Favaro et al., Mol Biotechnol 2013, 53: 74-9; Steiner et al., Eur J Neurol 2012, 19: 1278-91).

In patients with elevated intracranial pressure, lumbar puncture (LP) may be associated with the risk of brain herniation. Patients at high risk for this usually fatal complication include those who have a history of CNS disease (e.g., mass, lesion, or focal infection), new onset seizure within one week of presentation, papilledema, or focal neurological deficit. In these patients, a computed tomography (CT) scan prior to LP is recommended to identify hydrocephalus, mass lesions, cerebral edema, or midline brain shift. Additional diagnostic testing is determined by the season of the year, geographic location, travel history, recreational activities, occupational exposure, insect or animal contacts, vaccination history, and the patient's immune status.

Treatment of meningitis includes anti-infective and/or supportive therapy. When there is a suspicion of acute bacterial meningitis, anti-infective therapy should be initiated as soon as possible. Delays beyond three to six hours of hospital arrival are associated with increased morbidity and mortality (Chaudhuri et al., Eur J Neurol 2008, 15: 649-59; Lepur and Barsic, Infection 2007, 35: 225-31). The duration and choice of the anti-infective regimen is based on the specific pathogen and the presence of immunosuppression. Corticosteroids may be used to decrease inflammation (Brouwer et al., Cochrane Database Syst Rev 2013, 6: CD004405).

Reference: (Tunkel et al., Clin Infect Dis 2004; 39(9): 1267-1284).

Encephalitis: Encephalitis is an inflammation of the brain parenchyma associated with neurological dysfunction. Encephalitis is most frequently caused by viral pathogens, but may also be associated with bacteria, fungi, or parasites. Herpes simplex virus is the most common etiology. Clinical symptoms may include fever, headache, altered mental status, cognitive dysfunction, behavioral changes, focal neurological signs, and seizures.

Diagnostic evaluation of a patient with encephalitis includes a thorough history and physical examination. Diagnostic testing is dependent on the patient's history relating to season of the year, geography, travel, recreational activities, occupational exposure, insect or animal contacts, immunization history, and immune status. All patients with suspected encephalitis should have analysis of CSF unless contraindicated, complete blood count, renal and hepatic function tests, coagulation studies, and blood cultures. Imaging tests, such as magnetic resonance imaging (MRI) or CT of the brain and chest radiography, are also recommended. Based on clinical findings from physical examination, further work up may be warranted. Treatment of encephalitis includes anti-infective and supportive therapy (Solomon et al., J Infect 2012, 64: 347-73; Tunkel et al., Clin Infect Dis 2008, 47: 303-27).

Brain abscess: A brain abscess is a focal infection of the brain, commonly caused by a bacterial, fungal, or parasitic pathogen. It is often the result of an infection that spreads from the oropharynx, middle ear, or paranasal sinuses. Clinical symptoms may include headache, mental status changes, focal neurological deficits, a new onset seizure, nausea, vomiting, and fever. Delay in hospitalization and treatment, a low Glasgow coma score (GCS) on presentation, and presence of comorbidities are associated with an increased risk of death or permanent neurological deficit. Treatment of a brain abscess includes anti-infective therapy and often requires surgical drainage. In neurologically stable patients with small abscesses and an identifiable etiology, conservative treatment without surgical drainage may be appropriate. Conservative treatment would also be warranted when the risks of surgical intervention outweigh the benefits. In these patients, surgical drainage should be considered if the patient fails to improve or deteriorates (Arlotti et al., Int J Infect Dis 2010, 14 Suppl 4: S79-92; Brouwer et al., N Engl J Med 2014, 371: 447-56; Helweg-Larsen et al., BMC Infect Dis 2012, 12: 332; Honda and Warren, Infect Dis Clin North Am 2009, 23: 609-23).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Tunkel et al., Clin Infect Dis 2004; 39(9): 1267-1284; Tunkel et al., Clin Infect Dis 2008, 47: 303-27). The brain abscess content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:

● **Initial review, One:** ^(1, 2, 3, 4)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, Both:** ⁽⁵⁾

– Meningitis suspected

– Initial lumbar puncture inconclusive

● **ACUTE, ≥ One:** ⁽⁶⁾

– Brain abscess confirmed by head CT or brain MRI ⁽⁷⁾

● **Encephalitis actual or suspected and, All:** ⁽⁸⁾

– Head CT or brain MRI without evidence of cerebral edema ⁽⁹⁾

– Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽¹⁰⁾

● **Finding, ≥ One:**

– Temperature $\geq 100.4^{\circ}\text{F}(38.0^{\circ}\text{C})$ ⁽¹¹⁾

– WBC $\geq 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$

● **Meningitis and, One:** ⁽¹²⁾

● **Bacterial actual or suspected and, One:** ⁽¹³⁾

● **Cerebral spinal fluid (CSF) analysis and, ≥ One:** ⁽¹⁴⁾

– Glucose $< 66 \text{ mg/dL}(3.7 \text{ mmol/L})$

– Gram stain positive

– Neutrophils $> 2/\text{cu.mm}(2 \times 10^6/\text{L})$

– Polymerase chain reaction (PCR) positive ⁽¹⁵⁾

– Protein $> 60 \text{ mg/dL}(0.6 \text{ g/L})$

– WBC $> 10/\text{cu.mm}(10 \times 10^6/\text{L})$

● **Fungal, TB, or Lyme actual or suspected and, One:**

● **Cerebral spinal fluid (CSF) analysis and, One:** ⁽¹⁶⁾

– AFB smear positive

– Cryptococcal antigen or smear positive ⁽¹⁷⁾

– Glucose $< 66 \text{ mg/dL}(3.7 \text{ mmol/L})$

– Lymphocytes $> 2/\text{cu.mm}(2 \times 10^6/\text{L})$

– Polymerase chain reaction (PCR) positive ⁽¹⁸⁾

– Protein $> 60 \text{ mg/dL}(0.6 \text{ g/L})$

– WBC $> 10/\text{cu.mm}(10 \times 10^6/\text{L})$

● **Viral, actual or suspected, and, One:**

● **Cerebral spinal fluid (CSF) analysis and, ≥ One:** ⁽¹⁹⁾

● **Lab values, Both:**

– Protein $< 200 \text{ mg/dL}(2000 \text{ g/L})$

– WBC $> 5/\text{cu.mm}(5 \times 10^6/\text{L})$ and $< 1000/\text{cu.mm}(1000 \times 10^6/\text{L})$

– Polymerase chain reaction (PCR) positive ⁽²⁰⁾

● **INTERMEDIATE, ≥ One:** ⁽²¹⁾

– NIPPV ⁽²²⁾

– Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽²³⁾

● **CRITICAL, ≥ One:** ⁽²⁴⁾

● **Bacterial meningitis actual or suspected and, One:** ⁽¹³⁾

● **Cerebral spinal fluid (CSF) analysis and, ≥ One:** ⁽¹⁴⁾

– Glucose $< 40 \text{ mg/dL}(2.2 \text{ mmol/L})$

– Polymorphonuclear cells $> 80\%(0.80)$

– Protein $\geq 200 \text{ mg/dL}(2000 \text{ g/L})$

– WBC $\geq 1000/\text{cu.mm}(1000 \times 10^6/\text{L})$

● **Brain abscess and, Both:**

– Confirmed by head CT or brain MRI

● **Abscess drainage, One:**

- Scheduled or performed within 24h
- Contraindicated and mental status change ^(10, 25)
- Cerebral edema
- Coma, stupor, or obtundation or GCS ≤ 8
- DIC and blood product transfusion ⁽²⁶⁾
- ECMO or ECLS ^(27, 28)
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both:** ⁽²⁹⁾
 - Hypotension, $\geq$ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁰⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽³¹⁾
- Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(32, 33)
- Mechanical ventilation
- NIPPV and, $\geq$ **One:** ⁽²²⁾
 - Arterial $P_{O_2} \leq 39$ mmHg(5.2 kPa)
 - Arterial or venous $P_{CO_2} \geq 60$ mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 1, One:** ^(1, 3, 4)
(Symptom or finding within 24h)
- **OBSERVATION, One:** ⁽⁵⁾
 - Meningitis suspected and, **Both:**
 - Lumbar puncture, **Both:**
 - Initial lumbar puncture results inconclusive
 - Repeat planned within 12h
 - Neurological assessment at least 6x/24h ⁽³⁴⁾
- **ACUTE, $\geq$ One:** ⁽⁶⁾
 - Brain abscess and, **All:** ⁽⁷⁾
 - Confirmed by head CT or brain MRI
 - Neurological stability ⁽³⁵⁾
 - Abscess drainage not indicated
 - Anti-infective
 - Neurological assessment at least 6x/24h ⁽³⁴⁾
 - Encephalitis actual or suspected and, **All:** ⁽⁸⁾
 - Head CT or brain MRI without evidence of cerebral edema ⁽⁹⁾
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽¹⁰⁾
 - Finding, $\geq$ **One:**
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽¹¹⁾
 - WBC $\geq 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
 - Anti-infective ^(36, 37)
 - Neurological assessment at least 6x/24h ⁽³⁴⁾
 - Meningitis and, **One:** ⁽¹²⁾
 - Bacterial actual or suspected and, **All:** ⁽¹³⁾
 - Cerebral spinal fluid (CSF) analysis and, $\geq$ **One:** ⁽¹⁴⁾
 - Glucose < 66 mg/dL(3.7 mmol/L)
 - Gram stain positive
 - Neutrophils $> 2/\text{cu.mm}(2 \times 10^6/\text{L})$
 - Polymerase chain reaction (PCR) positive ⁽¹⁵⁾

- Protein > 60 mg/dL(0.6 g/L)
 - WBC > 10/cu.mm(10×10^6 /L)
- Anti-infective ⁽³⁸⁾
- Blood culture ⁽³⁹⁾
- Neurological assessment at least 6x/24h ⁽³⁴⁾
- Fungal, TB, or Lyme actual or suspected and, **All:**
 - Cerebral spinal fluid (CSF) analysis and, ≥ **One:** ⁽¹⁶⁾
 - AFB smear positive
 - Cryptococcal antigen or smear positive ⁽¹⁷⁾
 - Glucose < 66 mg/dL(3.7 mmol/L)
 - Lymphocytes > 2/cu.mm(2×10^6 /L)
 - Polymerase chain reaction (PCR) positive ⁽¹⁸⁾
 - Protein > 60 mg/dL(0.6 g/L)
 - WBC > 10/cu.mm(10×10^6 /L)
 - Anti-infective, ≥ **One:**
 - IV
 - Antifungal (includes PO) ⁽⁴⁰⁾
 - Anti-tuberculosis agent (includes PO) ⁽⁴¹⁾
 - Blood culture
 - Neurological assessment at least 6x/24h ⁽³⁴⁾
- Viral actual or suspected and, **All:**
 - Cerebral spinal fluid (CSF) analysis and, ≥ **One:** ⁽¹⁹⁾
 - Lab values, **Both:**
 - Protein < 200 mg/dL(2000 g/L)
 - WBC > 5/cu.mm(5×10^6 /L) and < 1000/cu.mm(1000×10^6 /L)
 - Polymerase chain reaction (PCR) positive ⁽²⁰⁾
 - Blood culture ⁽³⁹⁾
 - Neurological assessment at least 6x/24h ⁽³⁴⁾
- **INTERMEDIATE, ≥ One:** ⁽²¹⁾
 - NIPPV ⁽²²⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽²³⁾
- **CRITICAL, One:** ⁽²⁴⁾
 - CNS infection, ≥ **One:**
 - Bacterial meningitis actual or suspected and, **All:** ⁽¹³⁾
 - Cerebral spinal fluid (CSF) analysis and, ≥ **One:** ⁽¹⁴⁾
 - Glucose < 40 mg/dL(2.2 mmol/L)
 - Polymorphonuclear cells > 80%(0.80)
 - Protein ≥ 200 mg/dL(2000 g/L)
 - WBC ≥ 1000/cu.mm(1000×10^6 /L)
 - Blood culture ⁽³⁹⁾
 - Neurological assessment every 1-2h ⁽³⁴⁾
 - Brain abscess and, **All:**
 - Confirmed by head CT or brain MRI
 - Abscess drainage, **One:**
 - Scheduled or performed within 24h
 - Contraindicated and mental status change ^(10, 25)
 - Neurological assessment every 1-2h ⁽³⁴⁾
 - Anticonvulsant and, ≥ **One:**
 - Recurrent seizure ⁽⁴²⁾
 - Refractory seizure ⁽⁴²⁾
 - Cerebral edema and, ≥ **One:**
 - ICP monitoring (includes OR placement)
 - Corticosteroid

- Hyperosmolar therapy, ≥ **One:** ⁽⁴³⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
- Coma, stupor, or obtundation and neurological assessment every 1-2h ⁽³⁴⁾
- DIC and blood product transfusion ⁽²⁶⁾
- ECMO or ECLS ^(27, 28)
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** ⁽²⁹⁾
 - Hypotension, ≥ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Finding, ≥ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁰⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽³¹⁾
- Hyponatremia or SIADH, **Both:**
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(32, 33)
- Intervention, ≥ **One:**
 - Conivaptan or tolvaptan (includes PO) ⁽⁴⁴⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽⁴⁵⁾
- Mechanical ventilation
- NIPPV and, ≥ **One:** ⁽²²⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
- Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 2, **One:** ^(3, 4, 46)
 - OBSERVATION, **One:**
 - Meningitis, **One:**
 - Responder, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁷⁾
 - Afebrile
 - Repeat lumbar puncture negative
 - Pain controlled or manageable ⁽⁴⁸⁾
 - Partial responder, not clinically stable for discharge and requires continued stay, **One:**
 - Post lumbar puncture headache and, ≥ **One:**
 - Epidural blood patch performed within last 24h ⁽⁴⁹⁾
 - IV caffeine and fluid
 - ACUTE, ≥ **One:**
 - Brain abscess and, **Both:** ⁽⁷⁾
 - Anti-infective
 - Neurological assessment at least 3x/24h ⁽³⁴⁾
 - Encephalitis and, **Both:**
 - Anti-infective ⁽³⁶⁾
 - Neurological assessment at least 3x/24h ⁽³⁴⁾
 - Meningitis, **One:**
 - Bacterial, fungal, TB, or Lyme and, **Both:**
 - Anti-infective, ≥ **One:** ⁽⁵⁰⁾
 - IV
 - Antifungal (includes PO) ⁽⁴⁰⁾
 - Anti-tuberculosis agent (includes PO) ⁽⁴¹⁾

- Neurological assessment at least 3x/24h ⁽³⁴⁾
- **Viral and, Both:**
 - Culture or polymerase chain reaction (PCR) result pending ⁽²⁰⁾
 - Neurological assessment at least 3x/24h ⁽³⁴⁾
- **INTERMEDIATE, ≥ One:** ⁽⁵¹⁾
 - NIPPV ⁽²²⁾
 - **Neurological impairment or condition and, Both:** ⁽⁵²⁾
 - **Finding, ≥ One:**
 - Acute onset
 - Deterioration
 - Post Critical care
 - Neurological assessment every 3-4h, ≤ 2d ⁽³⁴⁾
 - Oxygen ≥ 40%(0.40) ≤ 2d ⁽²³⁾
 - **Seizure and, Both:**
 - Activity resolved
 - Anticonvulsant, continuous
- **CRITICAL, ≥ One:**
 - Brain abscess and neurological assessment every 1-2h ⁽³⁴⁾
 - **Cerebral edema and, ≥ One:**
 - Intracranial pressure monitoring (ICP) monitoring (includes OR placement)
 - Corticosteroid
 - **Hyperosmolar therapy, ≥ One:** ⁽⁴³⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
 - **Acute kidney injury (AKI) and, Both:** ⁽⁵³⁾
 - **Finding, ≥ One:**
 - **Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁵⁴⁾
 - Widening QRS
 - **Sodium, One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - **Intervention, ≥ One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁵⁵⁾
 - Diuretic, continuous
 - **Dialysis, One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - IV fluid resuscitation ≤ 2d ⁽⁵⁶⁾
 - CRRT ⁽⁵⁷⁾
 - **Anticonvulsant and, ≥ One:**
 - Recurrent seizure ⁽⁴²⁾
 - Refractory seizure ⁽⁴²⁾
 - DIC and blood product transfusion ⁽²⁶⁾
 - ECMO or ECLS ^(27, 28)
 - **Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, Both:** ⁽²⁹⁾
 - **Hypotension, ≥ One:**
 - MAP < 65 mmHg

- Systolic blood pressure < 90 mmHg and < baseline
- Finding, ≥ **One**:
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁰⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽³¹⁾
- Hyponatremia or SIADH and, **Both**:
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(32, 33)
- Intervention, ≥ **One**:
 - Conivaptan or tolvaptan (includes PO) ⁽⁴⁴⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽⁴⁵⁾
- Mechanical ventilation
- NIPPV and, ≥ **One**: ⁽²²⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁵⁸⁾
- Neurological impairment or condition and, **Both**: ⁽⁵²⁾
 - Finding, **One**:
 - Acute onset
 - Deterioration
 - Neurological assessment every 1-2h, ≤ 2d ⁽³⁴⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 3-7, **One**: ^(4, 59, 60)
- OBSERVATION, **One**:
 - Meningitis, **One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **Both**: ⁽⁴⁷⁾
 - Afebrile
 - Pain controlled or manageable ⁽⁴⁸⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One**:
 - For viral meningitis use Episode Day 3 at a higher level of care
 - For a condition other than meningitis use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
- ACUTE, **One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽⁴⁷⁾
 - Afebrile
 - WBC < 12,000/cu.mm(12x10⁹/L)
 - Bands ≤ 10%(0.10)
 - Infection resolving, **One**:
 - Viral meningitis suspected and bacterial etiology ruled out
 - Post acute anti-infective therapy arranged
 - Anti-infective therapy completed
 - Neurological stability ⁽⁶¹⁾
 - Pain controlled or manageable ⁽⁴⁸⁾
 - Tolerating PO or nutritional route established ^(62, 63)
 - Complication or comorbidity, **One**:
 - No complication or active comorbidity relevant to this episode of care ⁽⁶⁴⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One**:
 - Acute kidney injury (AKI) and, ≥ **One**:
 - Kidney function improving or within acceptable limits ⁽⁶⁵⁾
 - Outpatient dialysis regimen established
 - Adverse drug reaction resolving

- Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽⁶⁵⁾
- *C. difficile* infection and, **Both:** ⁽⁶⁶⁾
 - Decrease in frequency and severity of diarrhea
 - Tolerating PO ⁽⁶³⁾
- Functional impairment and, **One:** ⁽⁶⁷⁾
 - Discharge to home and, **One:** ⁽⁶⁸⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
- Sodium 130-150 mEq/L(130-150 mmol/L)
- Pneumonia and, **All:**
 - Afebrile
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽⁶⁵⁾
 - WBC < 12,000/cu.mm(12x10⁹/L)
 - WBC > 4,000/cu.mm(4 x10⁹/L)
 - Bands ≤ 10%(0.10)
- Seizures absent or controlled
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Brain abscess and, ≥ **One:**
 - Anti-infective regimen incomplete, ≥ **One:** ⁽³⁶⁾
 - Anti-infective ≤ 7d since initiation
 - Home IV anti-infective therapy not clinically appropriate and, ≥ **One:**
 - Afebrile ≤ 2d
 - Access to outpatient IV therapy unavailable
 - Caregiver unreliable or unable to provide care
 - Home environment unsafe
 - Infection unresolved and, **Both:**
 - Finding, ≥ **One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹¹⁾
 - WBC ≥ 12,000/cu.mm(12x10⁹/L)
 - Bands > 10%(0.10)
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 and, ≥ **One:** ⁽¹⁰⁾
 - Deterioration
 - Fluctuation
 - New onset
 - Focal neurological deficit ≤ 3d ⁽⁶⁹⁾
 - Anti-infective
- Encephalitis and, ≥ **One:**
 - Anti-infective regimen incomplete, ≥ **One:** ^(36, 37)
 - Anti-infective ≤ 3d since initiation
 - Herpes simplex viral encephalitis and anti-infective ≤ 10d since initiation
 - Home IV anti-infective therapy not clinically appropriate and, ≥ **One:**
 - Afebrile ≤ 2d
 - Access to outpatient IV therapy unavailable
 - Caregiver unreliable or unable to provide care
 - Home environment unsafe
- Infection unresolved and, **Both:**
 - Finding, ≥ **One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹¹⁾
 - WBC ≥ 12,000/cu.mm(12x10⁹/L)
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 and, ≥ **One:** ⁽¹⁰⁾
 - Deterioration
 - Fluctuation

- New onset
- Focal neurological deficit $\leq 3d$ ⁽⁶⁹⁾
- Anti-infective ^(36, 37)
- Meningitis, bacterial, fungal, TB, or Lyme and, $\geq$ **One:** ⁽⁷⁰⁾
 - Anti-infective regimen incomplete, **Both:**
 - Anti-infective $\leq 6d$ since initiation, $\geq$ **One:**
 - IV
 - Antifungal (includes PO) ⁽⁴⁰⁾
 - Anti-tuberculosis agent (includes PO) ⁽⁴¹⁾
 - Home IV anti-infective therapy not clinically appropriate and, $\geq$ **One:**
 - Afebrile $\leq 2d$
 - Access to outpatient IV therapy unavailable
 - Caregiver unreliable or unable to provide care
 - Home environment unsafe
 - Infection unresolved and, **Both:** ⁽⁵⁰⁾
 - Finding, $\geq$ **One:**
 - Temperature $\geq 100.4^{\circ}F(38.0^{\circ}C)$ ⁽¹¹⁾
 - WBC $\geq 12,000/cu.mm(12 \times 10^9/L)$
 - Bands $> 10\%(0.10)$
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 and, $\geq$ **One:** ⁽¹⁰⁾
 - Deterioration
 - Fluctuation
 - New onset
 - Focal neurological deficit $\leq 3d$ ⁽⁶⁹⁾
 - Anti-infective, $\geq$ **One:**
 - IV
 - Antifungal (includes PO) ⁽⁴⁰⁾
 - Anti-tuberculosis agent (includes PO) ⁽⁴¹⁾
 - Acute kidney injury (AKI) and, **Both:** ⁽⁵³⁾
 - Finding, $\geq$ **One:**
 - Urine output $< 0.5 mL/kg/h$
 - Creatinine, **One:**
 - $\geq 1.5x$ baseline and chronic kidney disease
 - $\geq 1.5x$ ULN ⁽⁷¹⁾
 - GFR $> 25\%(0.25)$ decrease from baseline ⁽⁷²⁾
 - Dialysis initiated
 - Intervention, $\geq$ **One:**
 - Volume expander $\leq 2d$ ^(73, 74)
 - Diuretic $\geq 2x/24h, \leq 3d$
 - Medication adjustment or discontinuation $\leq 3d$ and, $\geq$ **One:** ⁽⁷⁵⁾
 - Diuretic (includes PO)
 - Nephrotoxic agent (includes PO) ⁽⁷⁶⁾
 - Dialysis (initial) and $\leq 7d$ since initiation ⁽⁷⁷⁾
 - Dialysis discontinued requiring lab monitoring $\leq 2d$ since discontinuation ⁽⁷⁸⁾
 - Adverse drug reaction requiring change in medication $\leq 24h$ ⁽⁷⁹⁾
 - Blood glucose $\leq 24h$, **All:** ⁽⁸⁰⁾
 - Finding, $\geq$ **One:**
 - $> 250 mg/dL(13.9 mmol/L)$
 - $< 70 mg/dL(3.9 mmol/L)$
 - Blood glucose monitoring at least $4x/24h$
 - Insulin adjustment and, **One:** ^(80, 81)
 - IV $\geq 2x/24h$
 - SC $\geq 4x/24h$

- *C. difficile* infection, actual or suspected and, **Both:** ⁽⁸²⁾
 - Anti-infective (includes PO) $\leq 5d$ ^(83, 84)
 - Inadequate oral intake ⁽⁸⁵⁾
- Functional impairment (new) requiring rehabilitation initiation $\leq 24h$ and, $\geq$ **One:** ^(67, 86)
 - PT evaluation and training
 - OT evaluation and training
 - Speech-language evaluation and training
 - Cognitive evaluation and retraining
 - Swallowing evaluation and retraining
- Hyponatremia or SIADH, **Both:** ⁽⁸⁷⁾
 - Sodium 120-129 mEq/L(120-129 mmol/L)
 - Intervention, $\geq$ **One:**
 - Diuretic and oral sodium supplement
 - Fluid restriction
 - Medication administration, $\geq$ **One:**
 - Demeclocycline (includes PO)
 - Diuretic $\geq 2x/24h$
 - Sodium chloride 0.9%(0.009) ⁽⁴⁵⁾
- Inadequate oral intake and, $\geq$ **One:** ⁽⁸⁸⁾
 - IV fluid $\leq 4d$ ⁽⁸⁹⁾
 - Feeding tube, **Both:**
 - Insertion $\leq 24h$
 - Feedings initiated
- Pain uncontrolled and, **Both:**
 - Pain assessment ⁽⁹⁰⁾
 - Requiring change in medication, dose, or frequency ^(75, 91)
- Pneumonia confirmed by imaging and, **Both:** ⁽⁹²⁾
 - Finding, $\geq$ **One:**
 - Temperature $\geq 100.4^{\circ}F(38.0^{\circ}C)$ ⁽¹¹⁾
 - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽¹⁰⁾
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline ⁽⁷²⁾
 - WBC $\geq 12,000/cu.mm(12 \times 10^9/L)$
 - WBC $\leq 4,000/cu.mm(4 \times 10^9/L)$
 - Bands $> 10\%(0.10)$
 - Intervention, $\geq$ **One:** ⁽⁹³⁾
 - Requiring supplemental oxygen ⁽²³⁾
 - Anti-infective (includes PO)
 - Respiratory interventions at least $6x/24h$ ⁽⁵⁸⁾
- Post lumbar puncture headache and, $\geq$ **One:**
 - Epidural patch performed within last $24h$ ⁽⁴⁹⁾
 - IV caffeine and fluid $\leq 24h$
- Seizure, new onset and, $\geq$ **One:**
 - Initiation of anti-convulsant (includes PO) $\leq 2d$
 - Change in anti-convulsant medication or dose (includes PO) $\leq 2d$
- Post Critical care $\leq 24h$
- **INTERMEDIATE, $\geq$ One:**
 - NIPPV ⁽²²⁾
 - Neurological impairment or condition and, **Both:** ⁽⁵²⁾
 - Finding, $\geq$ **One:**
 - Acute onset
 - Deterioration
 - Neurological assessment every 3-4h, $\leq 2d$ ⁽³⁴⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2d$ ⁽²³⁾

- Seizure and, **Both:**
 - Activity resolved
 - Anticonvulsant, continuous
- **CRITICAL, ≥ One:**
 - Brain abscess and neurological assessment every 1-2h ⁽³⁴⁾
 - Cerebral edema and, ≥ **One:**
 - Intracranial pressure monitoring (ICP) monitoring (includes OR placement)
 - Corticosteroid
 - Hyperosmolar therapy, ≥ **One:** ⁽⁴³⁾
 - Mannitol
 - Hypertonic saline ≥ 3%(0.03) solution
- Acute kidney injury (AKI) and, **Both:** ⁽⁵³⁾
 - Finding, ≥ **One:**
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁵⁴⁾
 - Widening QRS
 - Sodium, **One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁵⁵⁾
 - Diuretic, continuous
 - Dialysis, **One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - IV fluid resuscitation ≤ 2d ⁽⁵⁶⁾
- CRRT ⁽⁵⁷⁾
- Anticonvulsant and, ≥ **One:**
 - Recurrent seizure ⁽⁴²⁾
 - Refractory seizure ⁽⁴²⁾
- DIC and blood product transfusion ⁽²⁶⁾
- ECMO or ECLS ^(27, 28)
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** ⁽²⁹⁾
 - Hypotension, ≥ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Finding, ≥ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁰⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽³¹⁾
- Hyponatremia or SIADH and, **Both:**
 - Sodium < 120 mEq/L(120 mmol/L) and symptomatic ^(32, 33)
 - Intervention, ≥ **One:**
 - Conivaptan or tolvaptan (includes PO) ⁽⁴⁴⁾
 - Diuretic ≥ 2x/24h
 - Fluid restriction
 - Sodium chloride 3%(0.03) ⁽⁴⁵⁾
- Mechanical ventilation

- NIPPV and, ≥ **One:** ⁽²²⁾
 - $\text{FiO}_2 > 50\%(0.50)$
 - Respiratory intervention every 1-2h ⁽⁵⁸⁾
- Neurological impairment or condition and, **Both:** ⁽⁵²⁾
 - Finding, **One:**
 - Acute onset
 - Deterioration
 - Neurological assessment every 1-2h, ≤ 2d ⁽³⁴⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 8, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁷⁾
 - Afebrile
 - $\text{WBC} < 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
 - Bands ≤ 10%(0.10)
 - Infection resolving and, **One:**
 - Post acute anti-infective therapy arranged
 - Anti-infective therapy completed
 - Neurological stability ⁽⁶¹⁾
 - Pain controlled or manageable ⁽⁴⁸⁾
 - Tolerating PO or nutritional route established ^(62, 63)
 - Complication or comorbidity, **One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁶⁴⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
 - Acute kidney injury (AKI) and, ≥ **One:**
 - Kidney function improving or within acceptable limits ⁽⁶⁵⁾
 - Outpatient dialysis regimen established
 - Adverse drug reaction resolving
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽⁶⁵⁾
 - *C. difficile* infection and, **Both:** ⁽⁶⁶⁾
 - Decrease in frequency and severity of diarrhea
 - Tolerating PO ⁽⁶³⁾
 - Functional impairment and, **One:** ⁽⁶⁷⁾
 - Discharge to home and, **One:** ⁽⁶⁸⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - Sodium 130-150 mEq/L(130-150 mmol/L)
 - Pneumonia and, **All:**
 - Afebrile
 - $\text{O}_2 \text{ sat} \geq 92\%(0.92)$ or within acceptable limits ⁽⁶⁵⁾
 - $\text{WBC} < 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
 - $\text{WBC} > 4,000/\text{cu.mm}(4 \times 10^9/\text{L})$
 - Bands ≤ 10%(0.10)
 - Seizures absent or controlled
 - **Non-responder**, not clinically stable for discharge and does not meet partial responder criteria, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review
 - **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge, **One:**
 - Does not meet partial responder criteria or at the Acute level of care, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review

- **CRITICAL, One:**

- **Non-responder**, not clinically stable for discharge, **One:**

- Does not meet partial responder criteria or at the Acute or Intermediate level of care, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, $\geq$ One: ⁽⁹⁴⁾

- **HOME, Both:**

- Level of care appropriateness, **Both:**

- Home environment safe and accessible ⁽⁹⁵⁾
 - Patient or caregiver demonstrates ability to manage care

- Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽⁹⁶⁾
 - Medication reconciliation ⁽⁹⁷⁾
 - Follow-up care planned ⁽⁹⁸⁾
 - Identify and address transportation needs

- **HOME CARE, Both:**

- Level of care appropriateness, **All:**

- Home environment safe and accessible ⁽⁹⁵⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(99, 100)

- Skilled services, $\geq$ **One:** ⁽¹⁰¹⁾

- Skilled nursing ⁽¹⁰²⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽¹⁰³⁾

- Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽⁹⁶⁾
 - Medication reconciliation ⁽⁹⁷⁾
 - Follow-up care planned ⁽⁹⁸⁾
 - Identify and address transportation needs

- **BEHAVIORAL HEALTH, Both:**

- Level of care appropriateness, **One:**

- Support systems available ⁽¹⁰⁴⁾

- Skilled treatment, $\geq$ **One:**

- Clinical assessment ⁽¹⁰⁵⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽¹⁰⁶⁾

- **SKILLED MEDICAL OR THERAPY, Both:**

- Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq$ 1x/wk
 - Treatment precluded in a lower level

- Services, $\geq$ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽¹⁰²⁾

- Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(107, 108)
 - Functional impairments requiring at least supervision ⁽¹⁰⁹⁾
 - Able to tolerate 1h/d of skilled therapy $\geq$ 5d/wk

- Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁹⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

- **SUBACUTE MEDICAL OR THERAPY, Both:**

- Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2 x/wk
 - Treatment precluded in a lower level
- Services, $\geq$ **One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽¹⁰²⁾
- Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(107, 108)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁰⁹⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5 d/wk and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁹⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3 x/wk
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(107, 108)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁰⁹⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5 d/wk and ≥ 2 therapy disciplines
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁹⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3 x/wk
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(107, 108)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁰⁹⁾
 - Able to tolerate ≥ 15 h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽¹¹⁰⁾
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁹⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽¹¹¹⁾
 - Ventilator dependent ≥ 6 h/d and weaning planned ^(112, 113)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽¹¹⁴⁾
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁹⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
056 DEGENERATIVE NERVOUS SYSTEM DISORDERS WITH MCC	n/a	6.1	CMS GMLOS
057 DEGENERATIVE NERVOUS SYSTEM DISORDERS WITHOUT MCC	n/a	4	CMS GMLOS
073 CRANIAL AND PERIPHERAL NERVE DISORDERS WITH MCC	n/a	3.9	CMS GMLOS
074 CRANIAL AND PERIPHERAL NERVE DISORDERS WITHOUT MCC	n/a	2.9	CMS GMLOS
075 VIRAL MENINGITIS WITH CC/MCC	n/a	6	CMS GMLOS
076 VIRAL MENINGITIS WITHOUT CC/MCC	n/a	3.2	CMS GMLOS
091 OTHER DISORDERS OF NERVOUS SYSTEM WITH MCC	n/a	4.5	CMS GMLOS
092 OTHER DISORDERS OF NERVOUS SYSTEM WITH CC	n/a	3.1	CMS GMLOS
093 OTHER DISORDERS OF NERVOUS SYSTEM WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
094 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITH MCC	n/a	8.6	CMS GMLOS
095 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITH CC	n/a	5.9	CMS GMLOS
096 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITHOUT CC/MCC	n/a	4.5	CMS GMLOS
097 NON-BACTERIAL INFECTION OF NERVOUS SYSTEM EXCEPT VIRAL MENINGITIS WITH MCC	n/a	9.1	CMS GMLOS
098 NON-BACTERIAL INFECTION OF NERVOUS SYSTEM EXCEPT VIRAL MENINGITIS WITH CC	n/a	6.1	CMS GMLOS
099 NON-BACTERIAL INFECTION OF NERVOUS SYSTEM EXCEPT VIRAL MENINGITIS WITHOUT CC/MCC	n/a	3.9	CMS GMLOS
865 VIRAL ILLNESS WITH MCC	n/a	4.3	CMS GMLOS
866 VIRAL ILLNESS WITHOUT MCC	n/a	2.8	CMS GMLOS
867 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH MCC	n/a	5.5	CMS GMLOS
868 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH CC	n/a	3.7	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
869 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
BRAIN ABSCESS	3%	9.2	InterQual
ENCEPHALITIS	8%	9.6	InterQual
MENINGITIS	17%	5.4	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Meningitis, expected progress within 72h :

- Vital signs improving
- Laboratory values normalizing
- Mental status improving
- Seizures absent or controlled
- Activity increasing, if appropriate
- Decreased supportive interventions (respiratory, continuous infusions) required

Consider:

- Physical therapy consult for ambulation and prevention of deconditioning
- Discharge planning

4:

Meningitis, care facilitation:

- Functional or ADL impairment - consider an alternate level of care for therapy needs
- Physical therapy provided in the home setting should be considered if the home is safe and accessible (Home Care)
- If home environment is unsafe, considered a skilled nursing facility (SNF)

Prolonged oxygen therapy expected

- Home oxygen therapy should be considered in a clinically stable patient who continues to require supplemental oxygen when there has been no change in oxygen requirements within last 2 days (Home)

Potential risk of readmission

- Knowledge deficit or non-adherence with condition management identified (Home Care, Readmission reduction team)
- Co-morbid chronic obstructive pulmonary disease or heart failure (Home Care, Readmission reduction team)

Prolonged ventilation and unable to wean

- Prolonged ventilator weaning expected - consider transfer to a LTAC for failed mechanical ventilation weaning

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, home homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies, i.e., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

7:

With modern technology, most brain abscesses are surgically drained (Brouwer et al., N Engl J Med 2014, 371: 447-56). Nonsurgical management of a brain abscess may be an option in carefully selected patients who meet the following criteria:

- Stable mental status, which may be indicated by a score of greater than 12 on the Glasgow Coma Scale (GCS)
- Abscess(es) less than 2.5 cm in diameter
- Identifiable etiology

Medically managed patients should receive serial imaging in order to assess the response to anti-infective therapy. Drainage of the abscess should be reconsidered in the setting of neurological deterioration, or if the abscess fails to resolve with treatment (Arlotti et al., Int J Infect Dis 2010, 14 Suppl 4: S79-92).

8:

Instruction: The Acute level of care is appropriate for patients with encephalitis without cerebral edema. The Critical level of care is required for patients with encephalitis and cerebral edema.

9:

Magnetic resonance imaging (MRI) is the most sensitive imaging study to evaluate patients with encephalitis. When MRI is unavailable or impractical, a computed tomography (CT) scan is recommended (Solomon et al., J Infect 2012, 64: 347-73; Tunkel et al., Clin Infect Dis 2008, 47: 303-27).

10:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

11:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

12:

In patients with acute bacterial meningitis, studies have demonstrated a benefit to a 4 day course of corticosteroids. Administration should either precede the first dose of the anti-infective or be given concurrently (Brouwer et al.,

Cochrane Database Syst Rev 2013, 6: CD004405). In patients with tuberculous meningitis, routine use of corticosteroids is recommended in patients who are negative for human immunodeficiency virus (HIV). However, evidence is insufficient to make a recommendation in HIV positive patients (Prasad and Singh, Cochrane Database Syst Rev 2008: CD002244).

13:

In addition to adhering to standard precautions of disease control, patients with *Neisseria meningitidis* meningitis should also be placed on droplet precautions. Post exposure chemoprophylaxis is also recommended for household contacts and health care workers exposed to respiratory secretions (Siegel et al., Am J Infect Control 2007, 35: S65-164; van de Beek et al., N Engl J Med 2006, 354: 44-53).

14:

These cerebrospinal fluid findings indicate a clinical suspicion of early bacterial meningitis (Mace, Emerg Med Clin North Am 2010, 28: 535-70; Cho and Venna, Neurol Clin 2010, 28: 1061-88).

15:

Although cerebrospinal fluid (CSF) culture remains the gold standard for confirming bacterial meningitis, polymerase chain reaction (PCR) testing is a useful diagnostic adjunct (van de Beek et al., Clin Microbiol Infect 2016, 22 Suppl 3: S37-62).

16:

The cerebrospinal fluid findings included in this criteria point, indicate a clinical suspicion of fungal, tuberculosis, or Lyme meningitis (Mace, Emerg Med Clin North Am 2010, 28: 535-70; Cho and Venna, Neurol Clin 2010, 28: 1061-88).

17:

Cryptococcal meningitis is caused by a fungus and is often diagnosed in immunocompromised patients. Recommended treatment varies according to the underlying immune status, but generally includes induction therapy with antifungal agents, followed by consolidation and maintenance phases (Makadzange and McHugh, Semin Neurol 2014, 34: 47-60; Perfect et al., Clin Infect Dis 2010, 50: 291-322).

18:

The effectiveness of polymerase chain reaction (PCR) testing in the setting of meningitis not caused by a bacterial or viral pathogen varies by the suspected causative organism. When available, PCR may be used as a diagnostic adjunct prompting further work-up (Steiner et al., Eur J Neurol 2012, 19: 1278-91).

19:

Cerebrospinal fluid (CSF) findings in patients with viral meningitis include elevated white blood cells with predominate lymphocytes, normal or mildly decreased glucose levels, and moderately elevated protein levels (Levin and Lyons, Am J Med 2018, 131: 25-32).

20:

Polymerase chain reaction (PCR) testing has high sensitivity and specificity in the diagnosis of viral infections of the central nervous system (Steiner et al., Eur J Neurol 2012, 19: 1278-91).

21:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

22:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application

of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

23:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(0.21)

1L 24%(0.24) 4L 36%(0.36)

2L 28%(0.28) 5L 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

24:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national

regulations.

25:

Brain abscess drainage may be contraindicated when the risks of surgical intervention outweigh the benefits, such as location at an inoperable or high risk anatomical site (Arlotti et al., *Int J Infect Dis* 2010, 14 Suppl 4: S79-92; Brouwer et al., *N Engl J Med* 2014, 371: 447-56).

26:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., *J Thromb Haemost* 2013:).

27:

Transition Plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan; 60%(0.60) of patients who are extracorporeal membrane oxygenation (ECMO) survivors are readmitted (Jen and Shew, *Pediatrics* 2010, 125: 1217-23).

28:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., *American College of Cardiology Perspectives and Analysis* 2018; Napp et al., *Herz* 2017, 42: 27-44; Guirand et al., *J Trauma Acute Care Surg* 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., *J Am Coll Cardiol* 2014, 63: 2769-78; Domico et al., *Pediatr Crit Care Med* 2012, 13: 16-21; MacLaren et al., *Intensive Care Med* 2012, 38: 210-20; Peek et al., *Health Technol Assess* 2010, 14: 1-46; Mugford et al., *Cochrane Database Syst Rev* 2008: CD001340).

29:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., *Crit Care Med* 2021, 49: e1063-e143; Rhodes et al., *Crit Care Med* 2017, 45: 486-552; Semler and Rice, *Clin Chest Med* 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., *Crit Care Med* 2021, 49: e1063-e143; Rhodes et al., *Crit Care Med* 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

30:

Patients with sepsis-induced tissue hypoperfusion can present with elevated blood lactate levels greater than 2 mmol/L(18.0 mg/dL) (Shankar-Hari et al., JAMA 2016, 315: 775-87).

31:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

32:

Symptoms of hyponatremia may include nausea, lethargy, headache, and muscle weakness. If hyponatremia worsens, symptoms may progress to mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, obtundation, and seizures should be reviewed at a higher level of care due to the need for more frequent neurological monitoring (Weismann et al., Endocr Connect 2016:).

33:

Symptoms of hyponatremia include:

- Coma
- Headache
- Lethargy
- Nausea
- Obtundation
- Seizures
- Stupor
- Vomiting

34:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

35:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

36:

Intravenous acyclovir should be initiated upon clinical suspicion of encephalitis based on preliminary diagnostic testing, or within 6 hours of hospital admission if initial results are unavailable. Other anti-infective agents should be initiated on the basis of specific epidemiologic or clinical factors. Patients who are suspected of having concomitant bacterial meningitis should receive antibiotic treatment. Additionally, when there is a high suspicion of rickettsial or ehrlichial infection, doxycycline is the agent of choice. Alternative agents, such as chloramphenicol, azithromycin, rifampin, and fluoroquinolones, may be administered on a case-by-case basis (Solomon et al., J Infect 2012, 64: 347-73; Tunkel et al., Clin Infect Dis 2008, 47: 303-27).

37:

Evidence is insufficient to recommend routine administration of corticosteroids as an adjunctive treatment of

encephalitis. Corticosteroids should be considered on a case-by-case basis depending on the causative organism and a review of the risk-benefit ratio (Solomon et al., J Infect 2012, 64: 347-73; Tunkel et al., Clin Infect Dis 2008, 47: 303-27).

38:

When there is a suspicion of acute bacterial meningitis, anti-infective therapy should be initiated as soon as possible. Delays beyond 3 to 6 hours of hospital arrival are associated with increased morbidity and mortality (Chaudhuri et al., Eur J Neurol 2008, 15: 649-59; Lepur and Barsic, Infection 2007, 35: 225-31).

39:

Guidelines recommend obtaining blood cultures, before the start of anti-infective therapy, once the diagnosis of bacterial meningitis is suspected, particularly when lumbar puncture (LP) is contraindicated because of the patient's condition or the cerebral spinal fluid (CSF) culture is negative (van de Beek et al., Clin Microbiol Infect 2016, 22 Suppl 3: S37-62; Tunkel et al., Clin Infect Dis 2004; 39(9): 1267-1284).

40:

First-line induction treatment of fungal meningitis includes at least one agent administered intravenously. For patients with intolerance to specific intravenous agents, alternative regimens consisting of oral formulations are indicated (Perfect et al., Clin Infect Dis 2010, 50: 291-322).

41:

First-line anti-infectives used to treat tuberculosis (TB) meningitis include isoniazid, rifampicin, pyrazinamide, and either streptomycin or ethambutol. Isoniazid is considered the most effective agent, and is only available in tablet form. In patients with multi-drug resistant TB meningitis, evidence supports the administration of fluoroquinolones as a component of the treatment regimen (Garg et al., Expert Rev Anti Infect Ther 2013, 11: 605-21). In addition, guidelines recommend adjunctive corticosteroid treatment for a 2 to 3 week duration (National Institute for Health and Clinical Excellence (NICE), Tuberculosis: clinical diagnosis and management of tuberculosis, and measures for its prevention and control, 2011).

42:

Refractory seizures, also known as uncontrolled, intractable, or drug-resistant seizures, are resistant to treatment and are poorly controlled with medication. Recurrent seizures are seizures that occur more than once and may occur without warning (Billington et al., J Clin Med 2016, 5:).

43:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., Neurocrit Care 2020, 32: 647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

44:

Fluid restriction should be avoided during the first 24 hours of therapy with conivaptan or tolvaptan as it may increase the likelihood of an overly rapid correction of serum sodium and may lead to the development of cerebral edema.

45:

Hypovolemic hyponatremia may be treated with sodium chloride 0.9% (0.009). For severe hyponatremia, treatment with an infusion of sodium chloride 3% (0.03) may be required. Rapid correction of serum sodium can cause spastic quadriparesis or quadriplegia, pseudobulbar palsy, dysarthria, mutism, pupillary or oculomotor abnormalities, stupor, coma, ataxia, dystonia, osmotic demyelinating syndrome, or even death (Weismann et al., Endocr Connect 2016:).

46:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights. For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

47:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

48:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

49:

Headache following lumbar puncture (LP) results from a persistent leak of cerebrospinal fluid at the site. The headache is typically worse when the patient is upright, and markedly relieved when supine. Most headaches will spontaneously resolve within a few days of the LP. Post LP headaches that do not resolve may benefit from placement of an epidural blood patch at the site of the leak, which usually relieves the symptoms (White, The lumbar epidural blood patch: A primer. 2019).

50:

In patients with bacterial meningitis, clinical condition is expected to improve within 48 hours after the initiation of anti-infective therapy. If no improvement is noted or the patient's condition deteriorates, the following factors may be considered: increased intracranial pressure (ICP) as a result of cerebral edema or hydrocephalus, cerebral thrombosis, inappropriate or ineffective choice of anti-infective, incorrect diagnosis, syndrome of inappropriate antidiuretic hormone secretion (SIADH), or superimposed infection (Chaudhuri et al., Eur J Neurol 2008, 15: 649-59). Repeat cerebrospinal fluid analysis is recommended if there is no clinical improvement after 48 hours (van de Beek et al., Lancet 2012, 380: 1693-702).

51:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

52:

Neurological deficit(s) may include:

- Altered level of consciousness
- Cognitive deficit
- Motor deficit
- Speech deficit
- Sensory deficit

Neurological disorder may include:

- Amyotrophic lateral sclerosis (ALS)
- Dementia (including Alzheimer's disease)
- Guillain-Barre syndrome (GBS)
- Multiple sclerosis (MS)
- Myasthenia gravis (MG)
- Parkinson's disease (PD)
- Transverse myelitis

53:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.

AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.

AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

54:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

55:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

56:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020).

The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zeng et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

57:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

58:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

59:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

60:

Meningitis, expected progress after 5d:

- Afebrile
- Vital signs normal
- WBC improving
- Activities at or close to baseline
- Preparing for discharge within the next 24h

If not, consider:

- Additional imaging to identify source of infection
- Additional culture and sensitivity to identify pathogen and assess for drug resistant organisms
- Antibiotic regimen change for persistent fever, leukocytosis, or worsening clinical status

61:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

62:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

63:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

64:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

65:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

66:

Signs indicating recovery from *Clostridioides difficile* include:

- Decrease in the severity and frequency of diarrhea
- Absence of fever
- Normal white blood cell count

Not all patients demonstrate fever or leukocytosis during the course of illness. Diet advancement should be based on improving clinical signs and symptoms.

67:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

68:

Functional impairments may be safely managed in the home environment when the patient is willing and able to

perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

69:

Neurological deficits include the following: aphasia (impaired comprehension or communication by speech, writing, or signs), ataxia (inability to coordinate voluntary muscle activity), dysarthria (disturbance of speech and language), muscle weakness, paresis (partial paralysis of voluntary muscles), paralysis (loss of power of voluntary muscle), visual field loss, or blindness.

Instruction: The severity of the deficit and the patient's hemodynamic and neurologic stability determine which level of care setting is appropriate.

70:

In patients with bacterial meningitis, complications are rare when there is a positive clinical response after 4 days of anti-infective treatment. Guidelines suggest hospital discharge may be appropriate if the patient receives at least 6 days of inpatient anti-infective therapy and the following conditions are met (Tunkel et al., Clin Infect Dis 2004; 39 (9): 1267-1284):

- Afebrile for at least 24 to 48 hours
- Clinical status stable or improving, without significant neurological impairment or seizure activity
- Adequate oral intake
- Access to home care for completion of anti-infective regimen and daily access to a physician
- Intravenous access device established if necessary
- Follow-up and emergency plans in place
- Patient and/or caregiver demonstrates compliance with treatment plan
- Discharge environment safe (e.g., access to telephone, utilities, food, and refrigerator)

71:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

72:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

73:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

74:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Pistolesi et al., J Nephrol 2018, 31: 797-812; Joannidis et al., Intensive Care Med 2017, 43: 730-49; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii-138).

75:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

76:

Nephrotoxic agents include nonsteroidal anti-inflammatory drugs (NSAIDs), aminoglycosides, amphotericin B, high doses of loop diuretics, and anti-viral agents or other drugs (e.g., acyclovir and foscarnet) (Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

77:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

78:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

79:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

80:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

81:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

82:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

83:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives. Guidelines describe multiple acceptable regimens but do indicate a preference for oral fidaxomicin for patients with mild to moderate disease. Alternative regimens are driven by disease severity and the number of episodes the patient has experienced. Treatment guidelines recommend discontinuation of the inciting anti-infective as soon as possible to prevent recurrence. Other interventions include IV fluids, electrolytes, or oral fluids to maintain hydration. Use of opiates and antidiarrheal medications should be avoided because decreased intestinal motility will exacerbate the infection (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

84:

Instruction: This criteria point is intended for anti-infectives initiated for the treatment of *C. difficile* infections.

The preferred treatment of *Clostridioides difficile* infections is based on the episode of illness and the severity of the symptoms. The Infectious Diseases Society of America (IDSA) recommends a variety of treatment regimens which are dependent on clinical presentation, including disease severity and how many episodes of disease the patient has experienced (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

85:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit (Scales, Br J Nurs 2014, 23).

86:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

87:

Hyponatremia (Seay et al., Am J Kidney Dis 2020, 75: 272-86)

88:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

89:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

90:

Pain assessment includes quality, intensity, location, frequency, duration, and exacerbating or alleviating factors. Uncontrolled pain should be monitored, assessed, and documented. Appropriate medication adjustment and other interventions should be taken to reduce pain.

91:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter

- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

92:

Instruction: This criterion includes aspiration pneumonitis. Aspiration is marked by the abnormal movement of oropharyngeal or gastric contents into the lower respiratory tract. Aspiration pneumonitis occurs when acidic gastric contents cause inflammation of the lung parenchyma. In the early stages, aspiration pneumonitis is not considered an infectious process as gastric acid prevents microbial growth. Anti-infective therapy should be considered if symptoms (e.g., shortness of breath, wheezing, hypoxemia) fail to resolve within 48 hours (Neill and Dean, Curr Opin Infect Dis 2019, 32: 152-7).

93:

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

94:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

95:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

96:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

97:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)

- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

98:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

99:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

100:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

101:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

102:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed

professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

103:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

104:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

105:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

106:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the

InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

107:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

108:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

109:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

110:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

111:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

112:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

113:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

114:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.